

Luis Tituana

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PROFESSIONAL SUMMARY

Electrical Engineer with a Ph.D. in Mechanical Engineering with 5+ years of hands-on experience in field robotics, control systems, optimization, and machine learning. Led development of autonomous agricultural robots and authored multiple peer-reviewed publications. Strong background in instrumentation, PLCs, electronics, and microcontrollers. Actively seeking full-time opportunities in intelligent systems, control, robotics, or automation.

EDUCATION

Ph.D in Mechanical Engineering

August 2020 – December 2025

University of Central Florida | Orlando, FL

M.S. in Electrical Engineering

August 2018 – May 2020

Southern Illinois University | Carbondale, IL (Fulbright Scholar)

B.S. in Electronics and Control

September 2008 – July 2015

National Polytechnic School | Quito, Ecuador

EXPERIENCE

University of Central Florida

Orlando, FL

Postdoctoral Scholar

January 2026 – Present

- Developed a Python-based AI-based subspace trajectory optimization software tool; managed codebase with Git for version control and iterative development.
- Provided technical guidance to three graduate students on research involving control systems, machine learning, and robotic systems performance.

Graduate Research Assistant

August 2020 – December 2025

- Developed and implemented a neural network-based suboptimal control algorithm and MPC framework for dynamical systems, including mathematical stability guarantees, in the Autonomous Robots and Controls Lab. (Mechanical and Aerospace Engineering Department).
- Led the integration and field deployment of two autonomous robotic harvesters, spanning sensing, embedded systems, motor control, and software validation. Improved end-effector success by 10% and increased harvesting rate by 24%.
- Provided technical leadership to a team of 4–5 engineers, coordinating subsystem testing, debugging, and performance improvements across the robotic harvester platform.
- Designed and built wiring harnesses, connectors, and power distribution, improving robustness and serviceability for field deployment; utilized SolidWorks for designing and 3D-printing robotic components.
- Developed embedded and controls software using Arduino, implementing encoder-based motor control, PID tuning, and communications via I2C and serial.
- Built data logging and performance evaluation frameworks in MATLAB, Python, and C++, leveraging DAQ tools and oscilloscope debugging to validate system behavior in field conditions.

Southern Illinois University

Carbondale, IL

Research Scholar

August 2018 – May 2020

- Designed and implemented a testbed for a magnetic manipulator in the Laboratory for Control and Optimization of the Electrical Engineering Department.
- Developed the control framework of the testbed using LabView and DAQ tools.
- Utilized SolidWorks for CAD design and 3D printing of the entire testbed for the magnetic manipulator.

National Polytechnic School

Quito, Ecuador

Lecturer and Laboratory Instructor

October 2015 – May 2018

- Presented lectures about Introductory Mathematics, Control II, Digital Systems, and Microprocessors.
- Tutored students in the Circuits Analysis, Power Systems and Digital Systems Laboratories.

Astronomic Observatory of Quito

Quito, Ecuador

Technical Assistant

November 2014 – September 2015

- Revised and maintained general security systems (CCTV, cameras and alarms).
- Automated and designed electronic control boards for the control of telescopes.

- Led and collaborated with in the design and implementation of the control system of a Quartz Atomic Clock.

National Polytechnic School

Quito, Ecuador

Physics Laboratory Instructor

June 2013 – July 2014

- Tutored students for Physics laboratory sessions with basic principles of physics.
- Maintained and repaired laboratory equipment used in Physics Laboratory experiments.

Microcircuits

Quito, Ecuador

Technical assistant

January 2013 – May 2013

- Designed and revised technical drawings for CCTV networks, security, alarms using AutoCAD.

Southern Regional Electric Company

Loja, Ecuador

Substations and Communications Department Assistant

May 2012 – August 2012

- Installed control panels for electric substations monitoring system with a SCADA system.
- Reviewed electrical drawings for local and remote-control systems of electrical substations.

SKILLS & INTERESTS

Languages: Python (Pytorch), C/C++, MATLAB, LabVIEW, PLCs (Ladder, FBD, SFC).

Technical: Control systems, Robotics, Raspberry Pi, Arduino, SolidWorks, ISIS Proteus, Electronics, Industrial instrumentation, Microcontrollers, Machine Learning.

Spoken languages: English (fluent), Spanish (native).

Interests: Basketball, salsa dancing, soccer.

PROJECTS

Small Autonomous Field Robot for Strawberry Harvesting

- Designed a control algorithm for robotic autonomous motion in an open field farm.
- Developed hardware for power distribution, signal conditioning, and actuator control.
- Redesigned end-effector grasping method in Solidworks and 3D-printed custom parts.
- Integrated embedded vision, sensors, and communication interfaces (I²C, UART).

Magnetic Manipulator

- Implemented a magnetic manipulator to steer magnetic particles with permanent magnet rings.
- Designed the main structure in Solidworks and 3D printed custom components for alignment and stability.
- Developed driver electronics and signal conditioning for sensor feedback and magnetic field control.
- Implemented LabVIEW-based data acquisition and control for real-time monitoring and system tuning.

Cartesian Robot for document classification

- Designed and implemented a Cartesian robot to classify documents based on a barcode.
- Developed embedded control electronics, PCB, power regulation circuits, and sensor signal conditioning.

Micromouse robot

- Designed and built a miniature robot to solve a labyrinth.
- Obtained second place in the Labyrinth category in International Robotics Competition "RGZL", Quito, 2015.

PUBLICATIONS

L. Tituaña, and Y. Xu, "Nonlinear robust suboptimal trajectory control in an Actor-Critic network based receding horizon control framework," *International Journal of Control*, 2025.

A. Rivera Palma, Y. Xu, **L. Tituaña**, and M. Fritts, "Scheduling of robotics or machinery operations in agricultural fields: A review," *2024 ASABE Annual International Meeting*, 2024.

L. Tituaña, A. Gholami, Z. He, Y. Xu, M. Karkee, R. Ehsani, "A Small Autonomous Field Robot for Strawberry Harvesting," *Smart Ag. Tech.*, pp. 100454, 2024.

L. Tituaña, Y. Xu, "Subspace Structured Neural Network for Rapid Trajectory Optimization," *IFAC-PapersOnLine*, vol. 56, no. 3, pp. 37-42, 2023.

N. Riahi, **L. Tituaña**, A. Komae, "Homotopy Continuation for Feedback Linearization of Noncontact Magnetic Manipulators," *2020 American Control Conf. (ACC)*, pp. 4295-4300, 2020.

C. Carabalí, **L. Tituaña**, J. Aguilar, O. Camacho, D. Chavez, "Inverse Response Systems Identification using Genetic Programming," *Proc. ICINCO 2017*, Spain, 2017.

REFERENCES

- **Yunjun Xu**
Professor - University of Central Florida
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- **Fernando Soria**
Associate - Exponent
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- **Andrew Fordon**
Aerospace Engineer - American Systems
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- **Arash Komae**
Associate Professor - Southern Illinois University
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